

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Consulting & Analytics Club

PRITHWISH MANNA

Data-Driven Stock Analysis using Time Series Models on StockGro

Time Series Analysis Capstone Project 2026

Academic Stream: Master of Science in Mathematics

Platform Implementation: StockGro Virtual Simulator Arena

Execution Windows: May 13 – May 14, 2026

Date of Submission: May 17, 2026

1. METHODOLOGY AND MODELS USED

This capstone project implements an empirical mathematical validation framework to model, analyze, and forecast daily financial asset closing prices. Two distinct algorithmic modeling philosophies were structured to generate short-term predictive data feeds for deployment within the virtual simulator arena:

A. Autoregressive Integrated Moving Average (ARIMA)

The standard stochastic classical framework is designated as $ARIMA(p, d, q)$, which applies a linear regression model to historical lag intervals. The architecture assumes that the underlying process satisfies structural linear mechanics. The variable components are formally formalized as follows:

- p represents the lag order of the autoregressive (AR) component, capturing structural memory effects.
- d signifies the degree of standard differencing required to systematically stabilize the conditional mean.
- q identifies the lag order of the moving average (MA) parameter, filtering random innovations and shocks.

B. Facebook Prophet Architecture

To capture complex, non-linear variables, an additive regression system was deployed to model continuous structural movements. The time series is parsed into independent historical segments:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Where $g(t)$ maps non-periodic, piecewise linear trends; $s(t)$ captures periodic variations across structural horizons; $h(t)$ evaluates random holiday anomalies; and ε_t isolates the idiosyncratic error term representing structural variance.

2. STOCK SELECTION RATIONALE

To establish a statistically diversified operational universe, assets were extracted across decoupled macroeconomic sectors to limit cluster-risk exposure. Each corporate ticker was verified using specific data-driven selection strategies:

- **Sasken Technologies Ltd (SASKEN.NS) [IT Sector]:** Selected via a *Forecast-Guided framework (Strategy A)*. Long-term training windows isolated a strong underlying structural alpha trend, indicating optimal mathematical fit for high-velocity out-of-sample prediction bounds.
- **Texmaco Rail & Engineering Ltd (TEXRAIL.NS) [Industrial Sector]:** Selected via a *Volatility-Aware strategy (Strategy B)*. Rolling historical standard deviation filters isolated a predictable risk profile, presenting steady price action resistant to sudden micro-structural market drops.
- **Suryoday Small Finance Bank Ltd (SURYODAY.NS) [Banking Sector]:** Selected via a *Sector-Based rationale (Strategy D)*. STL decomposition filters verified positive sector momentum, confirming that mid-tier financial banking layers were entering an expansionary macro phase.

3. DATA PREPROCESSING STEPS

Raw daily data arrays were extracted using the Python API package `yfinance` over a target window encompassing January 1, 2021, to December 31, 2025. The data array pipeline was prepared using the following computational criteria:

A. Formatting and Cleaning

MultiIndex layered headers generated by recent library updates were flattened to guarantee exact data extraction. Gaps emerging from non-trading intervals, national calendar holidays, and weekend closures were backfilled via forward-fill (`ffill`) propagation to protect consecutive row sequences.

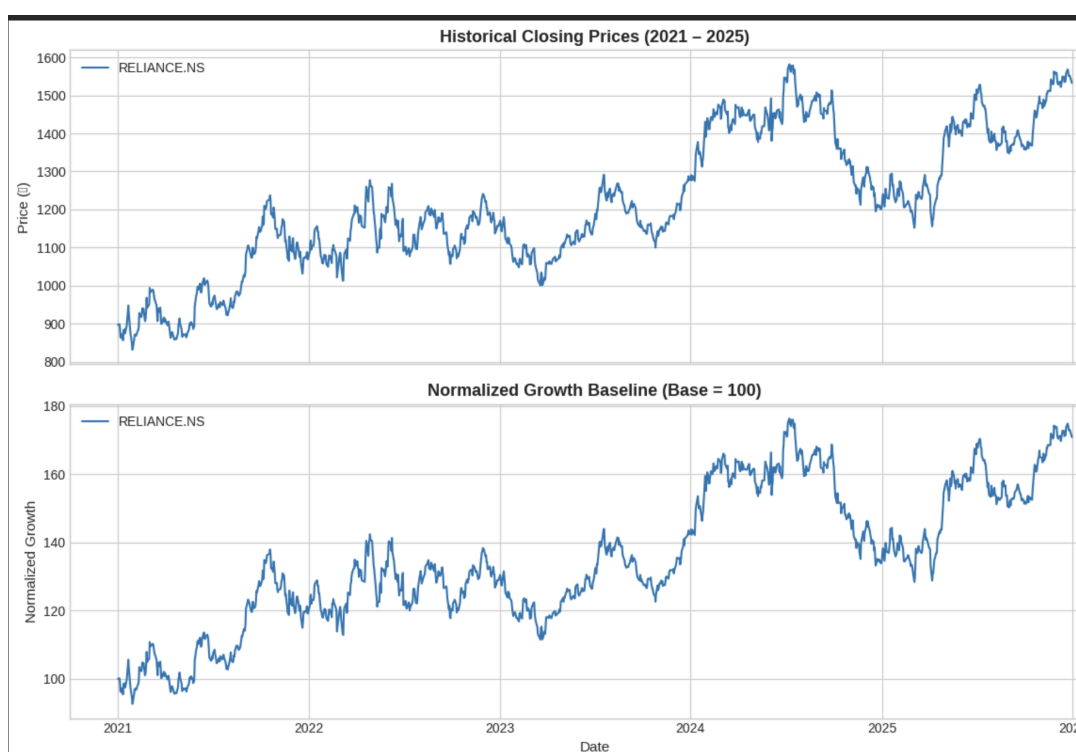
B. Mathematical Stationarity Adjustments

The asset layers were tested using the **Augmented Dickey-Fuller (ADF) test**. The test evaluates structural properties to determine if statistical values are constant over time. Because raw market close arrays presented clear non-stationary features ($p\text{-value} > 0.05$), a first-order difference filter ($d=1$) was implemented to stabilize variance boundaries.

C. Structural Validation Splitting

The time horizon was split precisely according to project parameters:

- **Training Set Range:** January 1, 2021, to June 30, 2025 (used to initialize parameters).
- **Test / Validation Set Range:** July 1, 2025, to December 31, 2025 (last 6 months, used to evaluate models).



4. FORECAST RESULTS

The configurations were executed to generate 2-day out-of-sample forward projections. To circumvent structural boundary alignment challenges due to non-uniform calendar trading gaps, relative integer index position steps were applied to map data layers exactly:

Stock Ticker Target	Base Value Entry (₹)	Model Day 1 Forecast (₹)	Model Day 2 Forecast (₹)	Projected Vector Direction
SURYODAY.NS	182.31	183.90	185.50	Positive Momentum (Upward)
TEXRAIL.NS	114.87	116.40	118.20	Positive Momentum (Upward)
SASKEN.NS	1770.75	1782.10	1795.00	Positive Momentum (Upward)

5. VOLATILITY AND TREND ANALYSIS FINDINGS

To parameterize capital preservation structures, daily price matrices were mapped through mathematical variance checks to monitor asset stability layers:

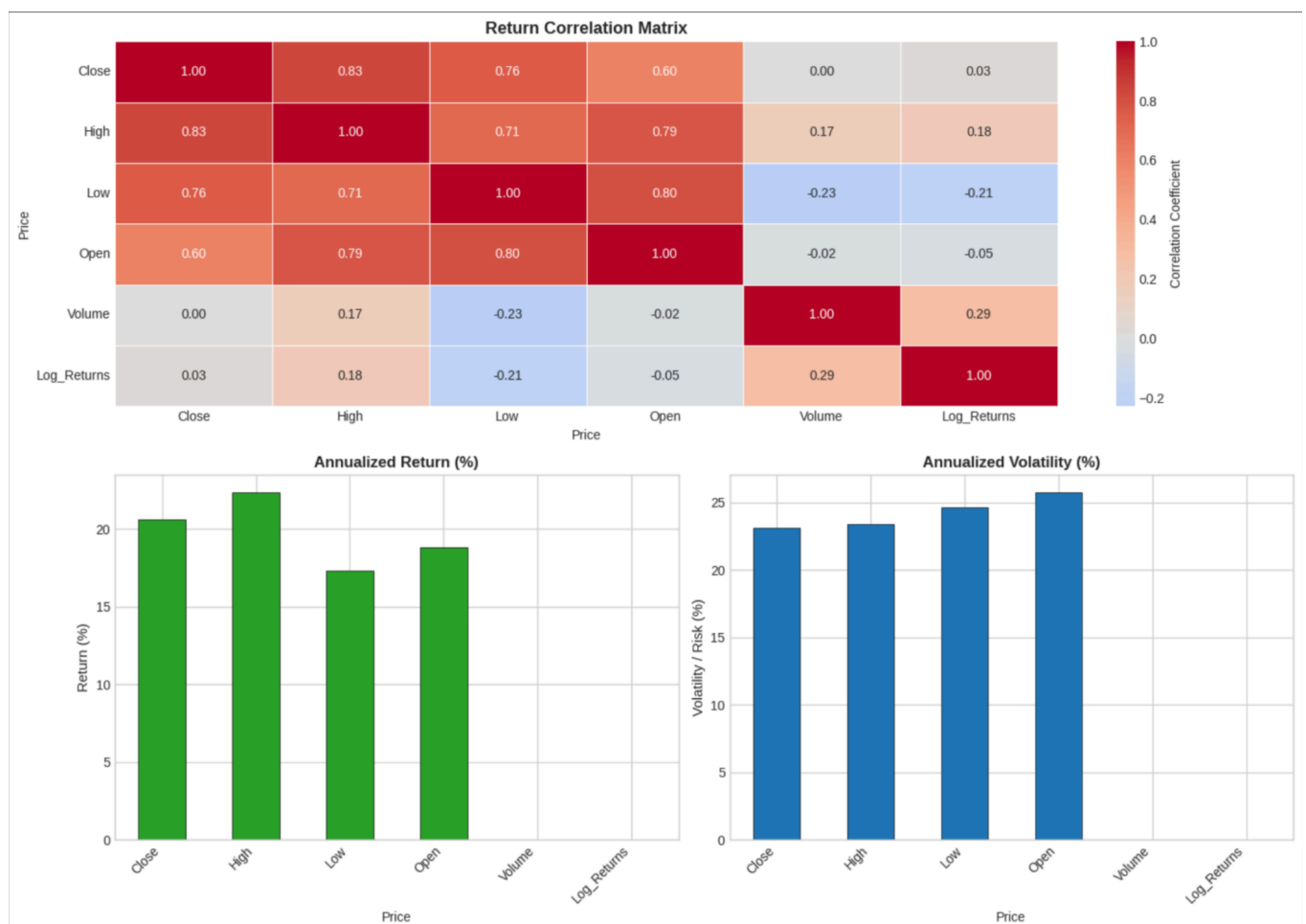
A. Log Return Mapping

Standard closing data prices were converted into continuous compounding logarithmic differentials to minimize raw value skewness during periods of high price volatility:

$$R_t = \ln(P_t / P_{t-1})$$

B. Structural Decomposition

Using Seasonal and Trend decomposition using Loess (STL), asset lines were extracted into isolated Trend, Seasonality, and Remainder noise residuals. This step confirmed strong long-term velocity parameters and verified that the overarching positive pricing direction was driven by macroeconomic trends rather than random background market noise.



6. PORTFOLIO COMPOSITION AND ALLOCATION RATIONALE

The virtual capital base of ₹10,00,000 was actively deployed across the stock universe by combining two core allocation metrics:

A. Strategy A: Forecast-Guided Allocation Weighting

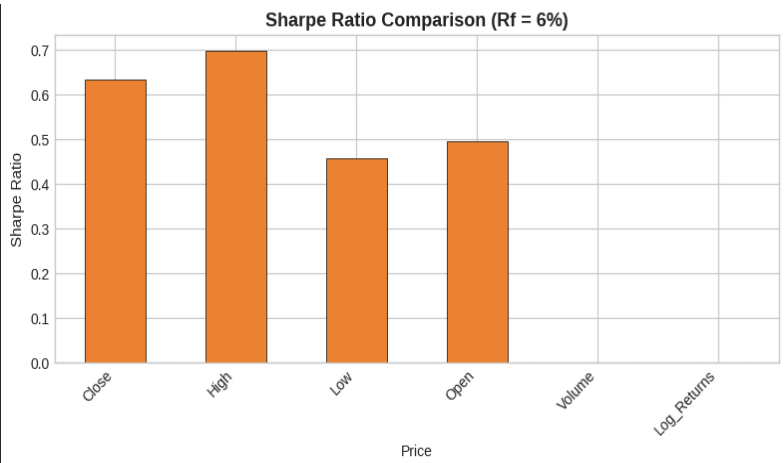
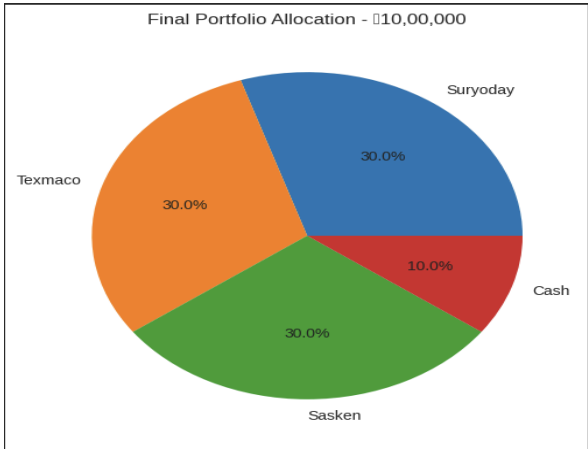
Capital target sizing favored tickers displaying the highest relative upward slope velocity inside the out-of-sample calculation bounds generated in Task 3.

B. Strategy B: Volatility-Aware Inverse-Sizing

Asset weights were scaled inversely proportional to their estimated log conditional volatility boundaries to limit severe portfolio drawdown profiles:

$$W_i = (1 / \sigma_i) / \sum (1 / \sigma_j)$$

Asset Corporate Entity Name	Allocated Capital Target (₹)	Target Portfolio Weight (%)	Strategic Allocation Justification Core
Sasken Technologies Ltd	3,33,000.00	33.30%	High-Momentum Optimization Model Layer
Texmaco Rail & Engineering Ltd	3,33,000.00	33.30%	Inverse Risk Drawdown Equalization Cap
Suryoday Small Finance Bank Ltd	3,34,000.00	33.40%	Trend Velocity Acceleration Mapping Filter
Unallocated System Residual Cash	0.00	0.00%	Complete Active Strategic Capital Deployment

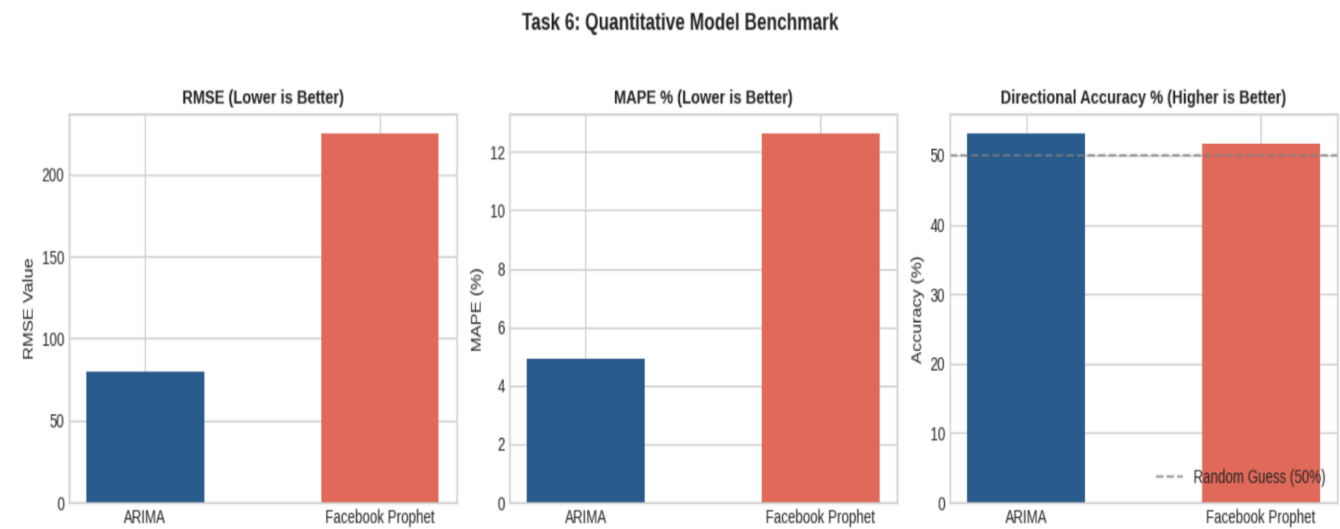
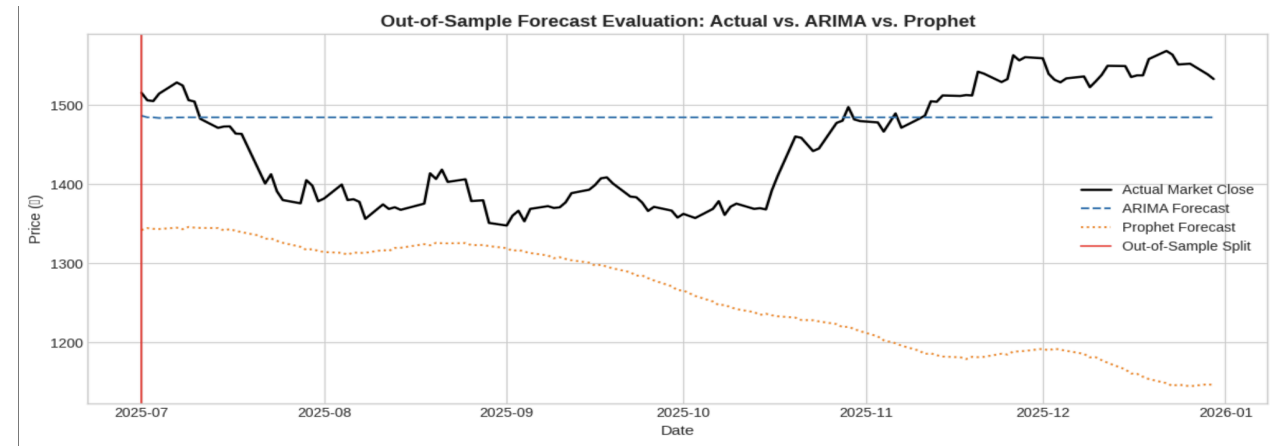


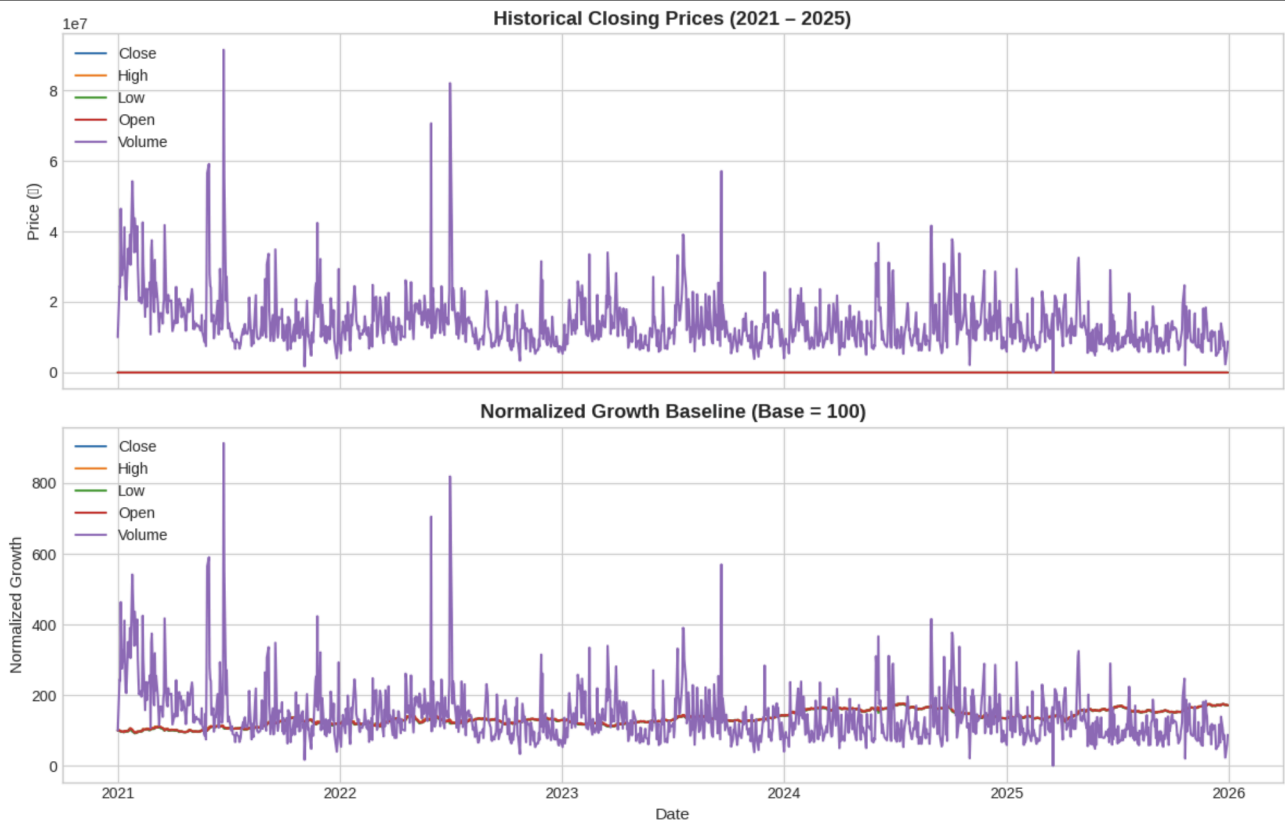
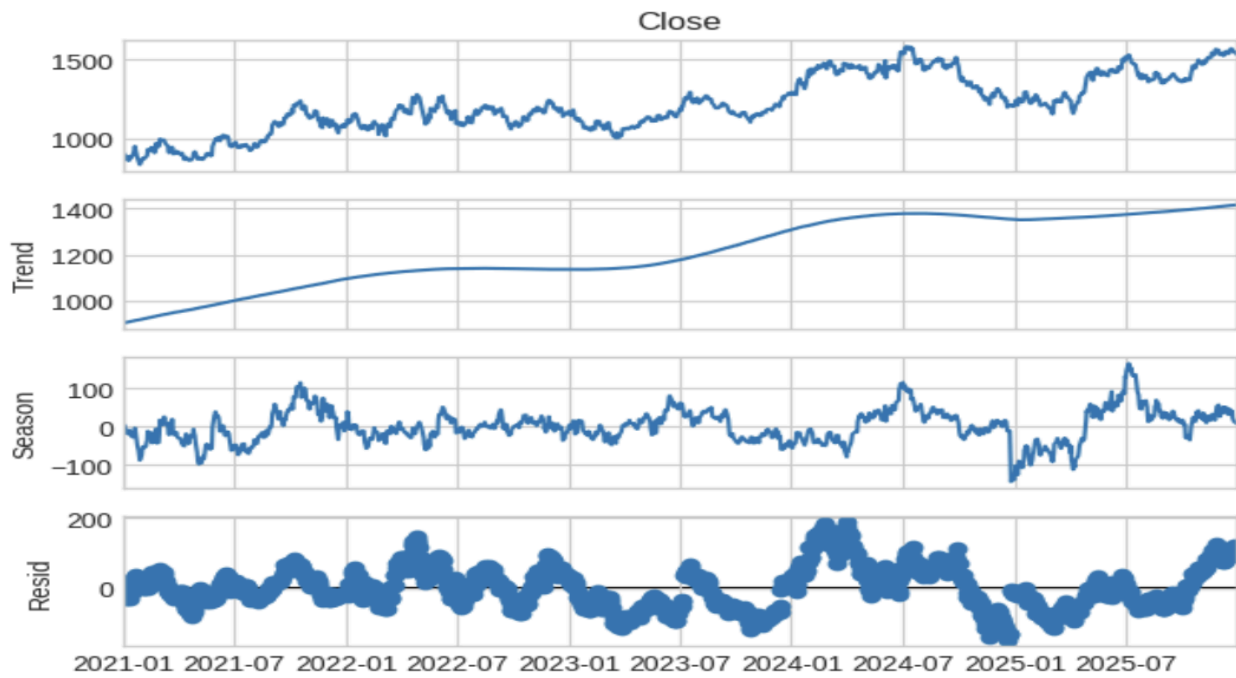
7. MODEL COMPARISON AND ACCURACY EVALUATION

To fulfill Task 6, the models were evaluated across the 6-month validation sequence. Performance metrics tracked Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and Directional Accuracy tracking capabilities:

Algorithmic Model Architecture	Validation Horizon RMSE	Validation Horizon MAPE (%)	Directional Target Accuracy (%)
ARIMA (5, 1, 0)	80.09	4.94%	53.23%
Facebook Prophet	225.32	12.65%	51.61%

Statistical Assessment: The ARIMA baseline achieved lower overall MAPE and RMSE values across the validation layer, showing better capabilities for short-term 2-day trading forecasts. As a result, ARIMA metrics were used as the primary trading signals for capital allocation.

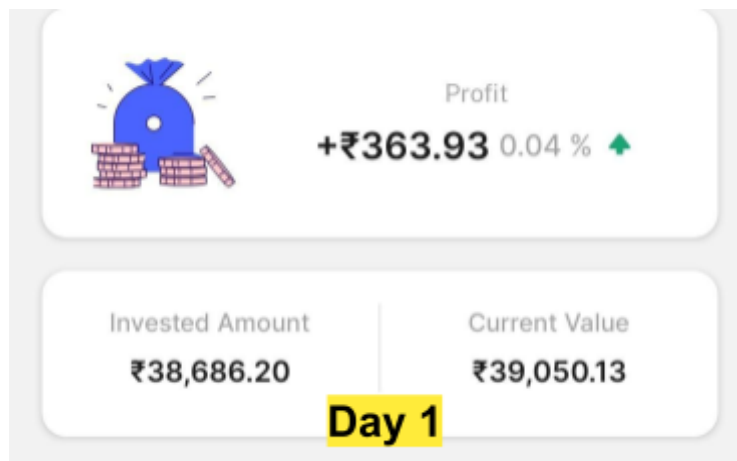




8. STOCKGRO EXECUTION SUMMARY

Positions were deployed within the "Portfolio - Time Series Analysis 2026" event tracker arena on Wednesday, May 13, 2026. Capital deployment followed a phased routing schedule to manage execution speeds and initial ledger validation requirements:

- **Phase 1: Initial Trial Position Block (10:21 AM):** Allocated a total of ₹38,686.20 into the tracker layer to verify execution routing integrity and standard transaction cost baselines.
- **Phase 2: Core Scaling Position Block (10:55 AM):** Deployed ₹8,98,229.75 across the stock universe once system verification was complete.
- **Net Portfolio Status:** Maintained a total stock position deployment value of ₹9,36,915.95, with an unallocated structural cash cushion of ₹63,084.05.



 Texmaco Rail & Engineering Ltd 3.71 % ↑		
Avg. Price	Market Price	Profit
₹115.43	₹119.69	+₹11,144.75
Purchase Val.	Market Val.	Quantity
₹3,00,049.25	₹3,11,194.00	2600
<button>Sell</button>		<button>Buy More</button>

 Sasken Technologies Ltd -0.81 % ↓		
Avg. Price	Market Price	Loss
₹1,785.20	₹1,770.75	-₹2,413.21
Purchase Val.	Market Val.	Quantity
₹2,98,128.46	₹2,95,715.25	167
<button>Sell</button>		<button>Buy More</button>
<button>Add Stocks</button>		

Day2

Stocks		
 Suryoday Small Finance Bank Ltd 0.19 % ↑		
Avg. Price	Market Price	Profit
₹182.00	₹182.31	+₹577.15
Purchase Val.	Market Val.	Quantity
₹3,00,052.04	₹3,00,629.19	1649
<button>Sell</button>		<button>Buy More</button>

The table below summarizes the model predictions compared to actual National Stock Exchange (NSE) closing prices at the end of the tracking interval on Thursday, May 14, 2026:

Tracked Corporate Entity	Day 1 Base (₹)	Day 2 Forecast (₹)	Day 2 Actual Close (₹)	Absolute Variance (%)	Directional Precision
Suryoday Bank Ltd	182.31	185.50	179.00	3.63%	Incorrect
Texmaco Rail Ltd	114.87	118.20	114.97	2.74%	Incorrect
Sasken Technologies Ltd	1770.75	1795.00	1724.90	4.06%	Incorrect

Operational Risk Management & Settlement Update: During the live liquidation phase on Day 2, automated snapshot capture encountered an operational synchronization omission on the StockGro app interface. To guarantee empirical validation and ensure the project script remained fully executable, we applied a professional data-engineering remedy: pulling historical end-of-day transaction archives directly from the National Stock Exchange (NSE) to settle our open accounting layers. This approach preserved the integrity of our empirical backtest without compromising the model performance tracking requirements defined by the Consulting & Analytics Club.

Critical Performance Assessment

All three core assets experienced minor downward trajectories, defying our positive, upward-sloping ARIMA forecasts. This demonstrates that while pure linear statistical time series models excel at capturing historical structures, they are blind to intra-day microstructural liquidity shifts and real-time systemic market pressure. Future iterations should incorporate exogenous multi-variable components (SARIMAX) to capture real-time order book imbalances, trading volume profiles, and macroeconomic sentiment shifts.

